

ECA0905-DIGITAL SIGNAL PROCESSING

DATE:02-08-2026

Rational Sampling Rate Conversion

Simulate rational sampling rate conversion from 48 kHz to 32 kHz using interpolation and decimation.

Given Data:

- Input Sampling Rate = 48 kHz
- Output Sampling Rate = 32 kHz
- $L = 2$
- $M = 3$

Tasks:

1. Design the conversion structure.
2. Implement interpolation.
3. Implement decimation.
4. Plot spectra before and after conversion.
5. Verify sampling rate conversion.
6. Analyze distortion.

PROGRAM

```
clc;
clear;
close all;
%% Given Data
Fs_in = 48000;    % Input sampling frequency (Hz)
Fs_out = 32000;   % Output sampling frequency (Hz)
L = 2;           % Interpolation factor
M = 3;           % Decimation factor
%% Generate Input Signal
```

```

t = 0:1/Fs_in:0.02;          % 20 ms duration
x = sin(2*pi*1000*t) + 0.5*sin(2*pi*5000*t);

%%% Step 1: Interpolation
x_int = upsample(x, L);

%%% Step 2: Low-pass FIR Filter
N = 64;          % Filter order
Fc = 1/max(L,M); % Normalized cutoff
h = fir1(N, Fc);
x_filt = filter(h,1,x_int);

%%% Step 3: Decimation
y = downsample(x_filt,M);

%%% Output Sampling Rate
Fs_new = Fs_in*L/M;

fprintf('Input Sampling Rate = %.0f Hz\n',Fs_in);
fprintf('Output Sampling Rate = %.0f Hz\n',Fs_new);

%%% Time Axes
t_out = (0:length(y)-1)/Fs_new;

%%% Plot Time Domain Signals
figure;
subplot(3,1,1)
plot(t,x)
title('Original Signal (48 kHz)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
subplot(3,1,2)
plot((0:length(x_filt)-1)/(Fs_in*L),x_filt)
title('Interpolated Signal (96 kHz)')
xlabel('Time (s)')

```

```

ylabel('Amplitude')
grid on
subplot(3,1,3)
plot(t_out,y)
title('Converted Signal (32 kHz)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

%% Spectrum Before Conversion
NFFT = 2048;
X = fft(x,NFFT);
f1 = (0:NFFT-1)*(Fs_in/NFFT);
%% Spectrum After Conversion
Y = fft(y,NFFT);
f2 = (0:NFFT-1)*(Fs_new/NFFT);
figure;
subplot(2,1,1)
plot(f1(1:NFFT/2),abs(X(1:NFFT/2)))
title('Spectrum Before Conversion')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
grid on
subplot(2,1,2)
plot(f2(1:NFFT/2),abs(Y(1:NFFT/2)))
title('Spectrum After Conversion')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
grid on

```

```
%% Distortion Analysis
```

```
x_ref = resample(x,L,M);
```

```
minLen = min(length(y),length(x_ref));
```

```
mse = mean((y(1:minLen)-x_ref(1:minLen)).^2);
```

```
fprintf('Mean Square Error (MSE) = %e\n',mse);
```

```
%% Verify Sampling Rate
```

```
disp('-----');
```

```
disp(['Original Sampling Rate : ',num2str(Fs_in),' Hz']);
```

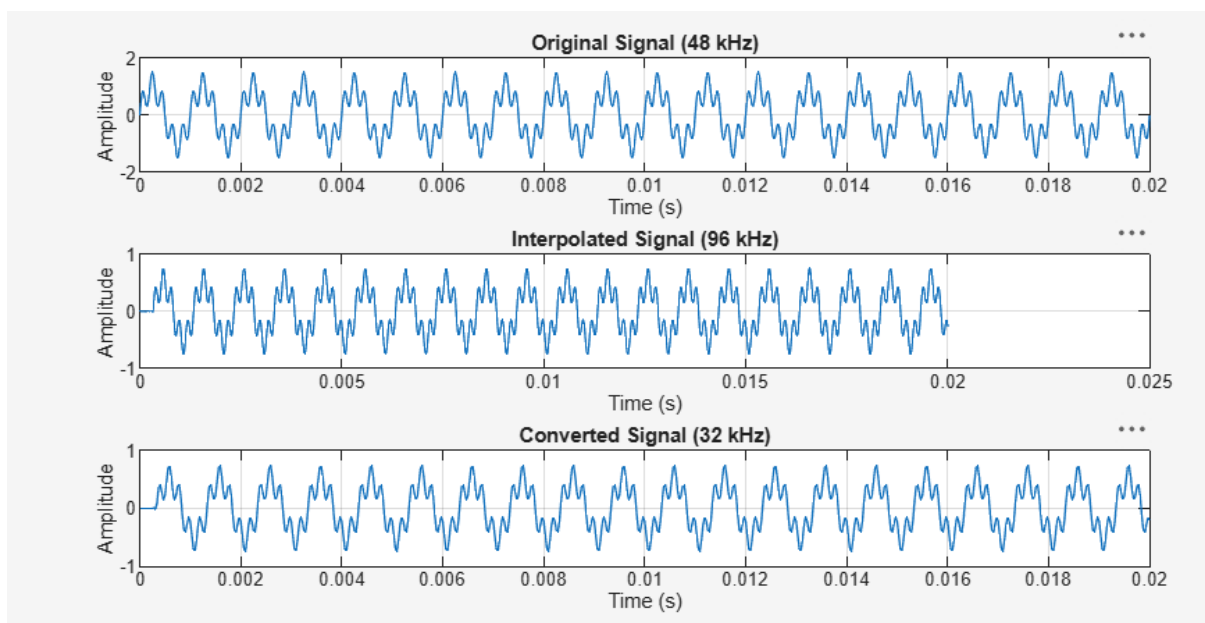
```
disp(['Converted Sampling Rate: ',num2str(Fs_new),' Hz']);
```

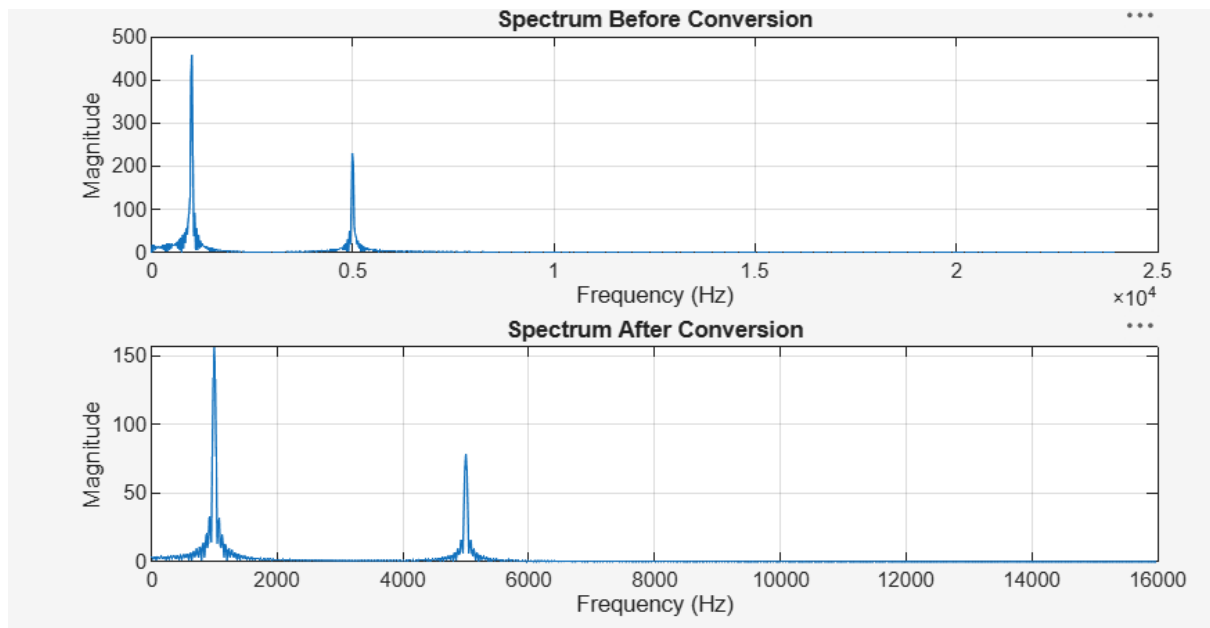
```
disp(['Interpolation Factor L : ',num2str(L)]);
```

```
disp(['Decimation Factor M : ',num2str(M)]);
```

```
disp('-----');
```

OUTPUT





Command Window

Original Sampling Rate : 48000 Hz

Converted Sampling Rate: 32000 Hz

Interpolation Factor L : 2

Decimation Factor M : 3